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Potential of new 250-nautical mile concentric circle allocation system for improving the donor/recipient HLA matching: Development of new matching algorithm

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ABSTRACT

Background: High-resolution typing of human leukocyte antigen (HLA) may revolutionize the field of kidney transplantation by selection of low immunogenic grafts. The new 250-nautical mile circle allocation system offers a unique opportunity to find low HLA immunogenic donors for eligible recipients.
Methods: 501 transplant candidates from the University of Toledo Medical Center (UTMC) between 2015 and 2019, registered at the Scientific Registry of Transplant Recipients (SRTR) were virtually matched to 4812 donors procured within 250-nautical miles using an in-house-developed simulation algorithm. Immunogenicity of HMS (hydrophobic mismatch score) ≤10 was measured based on imputed high-resolution HLAs. Simulated "optimal" matches with a KDPI≤50 % were compared with the transplant cohort between 2000 and 2010 with their kidney allograft survivals.
Results: Out of 501 recipients 500 (99.8 %) were matched with donors ≤10 HMS and KDPI ≤50 %. The average HMS value for simulated transplants was 1.4 (range 0–10) versus 6.3 (range 0–75) in the retrospective cohort ($p < 0.001$). The simulated model had a median mismatch number of 3/6, while the reference cohort 4/6 among HLA-A/B/DR antigens ($p < 0.001$). The estimated median graft survival was 18.2 years for the simulated cohort vs. 13.4 years in the real-life cohort ($p < 0.001$), gaining 4.9 years per transplant and 2450 survival years for all patients. For year 2014, out of 98 patients and 659 donors, each recipient had a median number of 141 donors (HMS < 10; range 8–378). Similar values were found for patients between 2015 and 2019.
Conclusion: Donors within 250-nautical miles proffers excellent and multiple options for finding well-matched low immunogenic HLA kidney donors for UTMC patients, thus significantly improving their chances for long-term allograft survival.

1. Introduction

Despite the overwhelming advantage of kidney transplantation over conservative renal replacement therapy options, both for patients and the national healthcare systems [1,2], all needed kidney transplants are not performed due to the lack of kidney donors [3]. Patients with end-stage renal disease are therefore placed on the waiting list for kidney transplantation upon evaluation. The median waiting time nationwide for candidates for the first-time kidney transplantation (listed in 2011–2014) was 3 years, and in March 2024 the number of patients on the kidney waiting list exceeded 89 thousand [3]. Contributing factor to this crisis is the fact that most kidney transplants are poorly matched at human leukocyte antigens (HLAs) [3–5]. The average kidney allograft survival for deceased donors is only around 10 years with a median of 4 out of 6 mismatched HLA-A/B/DR antigens [6].

The nationwide waiting time for kidney transplants ranges from 1 year to 8 years as it is influenced by multiple factors, such as transplant center offer acceptance policies as well as allocation algorithms guiding offer of procured kidneys. The somewhat arbitrary behavior of individual transplant centers has unintentional, but measurable national

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impact, therefore contributing to unequal distribution of kidney transplants throughout various donor service areas [7]. Geographic disparities have been persistent since introduction of the new Kidney Allocation System (KAS) in 2014, which intended to improve access to transplants for highly sensitized patients [8]. In attempt to reduce mostly geographic variations in waiting times for transplantation, the concentric 250-nautical-mile circle allocation policy was introduced in 2018. With this policy, the donation service area become extended to a 250-nautical-mile radius from a given transplant center. The complexity in implementation resulted in unintended and unfavorable consequences for kidney transplant candidates, such as increased kidney discard rates increased to 20 % and cold ischemia time extended to an average of 25 h [9]. It remains to be seen whether this update to the allocation policy would affect access to the HLA-matched donors. We propose that sudden accessibility to multiple potential donors offer an unprecedent opportunity to re-examine the allocation policy to take advantage of huge pool of HLA diversity among large number of potential donors for each potential transplant candidate.

The benefit of HLA matching for kidney transplantation has been shown repeatedly [10–12] and was considered in previous iterations of UNOS policies. However, requirements for HLA matching had to be relaxed due to the increase in waiting times for certain candidates, especially patients of African American race with highly diverse HLA antigens [7]. We recently proposed that this restriction is only the case when the traditional antigenic HLA-A/B/DR matching is used, whereas the use of alternative HLA immunogenicity scales based on HLA amino acid (a.a.) sequence would allow for much more flexible matching of donors to patients [13]. This is because continuous nature of such a.a.-based scales allows to calculate electrostatic (EMS), hydrophobic (HMS), and a.a. (AMS) mismatch scores, thus providing for more freedom and flexibility in the evaluation of HLA choice of donors considering that the scale may be from 0 (perfect match all a.a.) to 100 (mismatch at multiple a.a.). In addition, individual a.a. disparities may have different contribution to the graft immunogenicity making the measurements more versatile and precise [14]. We focused our previous work on the physicochemical immunogenicity algorithm developed by the group of Kosmoliaptsis et al. [15,16]. The continuous HMS immunogenicity scale defined cohorts of patients with low to high immunogenic scores: HMS $= 0$ had an average kidney graft survival of 18.1 years; HMS ≤ 3 of 16.7 years; HMS ≤ 7.8 of 14.3 years; and HMS > 7.8 of 12.5 years [6]. Similar flexible HMS scale allowed us to find excellent low immunogenic HMS matching for African American (AA) patients without worsening matches for other races [13]. Among best matched AA patients (HMS $=$ 0) graft survival was 16.1 years and well-matched grafts (HMS ≤ 3) was 14.0 years. The actual median graft survival among AA recipients was 12.0 years [13]. Application of high-resolution HLA typing methods and HMS-based matching may offer unique opportunity to improve HLA matching for patients with deceased donors [6,13].

Since every donor is evaluated for kidney donor profile index (KPDI), we also included this requirement for our analysis [17]. The KPDI evaluates the likelihood of graft failure based on 10 donor factors into a single number [age, ethnicity, hypertension, diabetes, cause of death, height, weight, donor type (deceased or BD) and HCV status] [18]. Based on previous analysis KPDI $\leq$20 % represent low risk, KPDI between 21 and 79 % represent standard risk, and KPDI $>$80 % represents high risk for poor outcomes [19]. Out of 4006 deceased donor kidney transplant recipients, 451 (11 %) recipients experienced allograft loss within 3 years. Compared to those with a KDPI of 0–25 %, recipients who had a KDPI $>$75 % experienced a 2-fold increased risk of 3-year allograft loss with adjusted hazard ratio (HR) of 2.04 [95 % confidence interval (CI) 1.53–2.71]. The adjusted HR for kidneys with a KDPI of 26–50 % was 1.27 (95 % CI 0.94–1.71) and for 51–75 % was 1.31 (95 % CI 0.96–1.77) [20]. Considering its importance, we factored the KPDI in our updated allocation simulation.

Herein, we explored the idea for quantifying chances of favorable HLA matching for candidates in one transplant center whose donor pool would be withing the 250-nautical-mile radius. We matched our patients at the University of Toledo Medical Center (UTMC) to donors registered within 250-nautical miles from UTMC. We run their allocation simulation to estimate the probability of getting a well-matched HLA transplant for each transplant candidate and assessed the resulting cohort for their immunogenicity and prospective graft survival. In addition, we performed an analysis showing that each recipient may have multiple possible donors matching strict requirements for low HLA immunogenic transplants with equal or lower HPDI than 50 %.

2. Materials and methods

2.1. SRTR database

This study used data from the Scientific Registry of Transplant Recipients (SRTR). The SRTR data system includes data on all donor, waitlisted candidates, and transplant recipients in the US, submitted by the members of the Organ Procurement and Transplantation Network (OPTN). The Health Resources and Services Administration (HRSA), U. S. Department of Health and Human Services provides oversight to the activities of the OPTN and SRTR contractors.

2.2. Patient, donor and reference cohorts

Data about kidney transplant candidates registered at the University of Toledo Medical Center between January 1, 2014 and December 31, 2019, were mined from the CAND_KIPA file of the SRTR database. Donor data were extracted from the DONOR_DECEASED file. Donors procured by OPOs within 250-nautical-mile radius from UTMC with recovery dates between 1/1/2014 and 12/31/2019. Demographic information was used for the candidate and donor cohorts.

Reference cohort was used for comparison of matching results and graft survival estimates between the simulated transplants and the actual retrospective data. We used the nationwide patient data from SRTR described before [21]. The cohort included adult transplant recipients transplanted between 2000 and 2010 and had known transplantation outcome data with follow-up time of at least 9 years.

2.3. HLA immunogenicity and allocation simulation algorithm

We used an optimized version of the previously published allocation simulation algorithm [22]. Briefly, selected cohorts of donors and recipients described above were used as inputs, and random donor-recipient pairs were generated observing the requirement of ABO compatibility. In the resulting transplants, physicochemical HLA immunogenicity was calculated. This process started with the imputation of high-resolution HLA types for both donor and recipients using frequency tables of HLA haplotypes among patient-reported races, as in the HaploStats algorithm [23]. Imputed high-resolution HLA types were then converted to amino acid sequences in HLA protein, and both in inter- and intra-locus comparisons between donor and recipient in each amino acid residue were made fashion. Mismatched amino acids were used for the calculation of final electrochemical (EMS), hydrophobic (HMS) or amino acid sequence (AAMS) mismatch scores. Since all three scores are interrelated [24], results were based on HMS scores are shown for brevity.

To address the impact of donor clinical confounders on the allocation simulation, we calculated KDPI values for each donor based on the following donor variables: age, BMI, race, history of diabetes, history of hypertension, cause of death (cerebrovascular incident or not), and donation after cardiac death status (yes or no). An additional allocation simulation was run where KDPI value of 50 % was set as an acceptability threshold and transplants with KDPI value greater than 50 % were re-matched.

2.4. Survival statistical analyses

Survival Analysis analyses time-to-event data and its objective is to model and examine data where outcome variable is the time until the occurrence of an event of interest. The primary endpoint in survival analyses was graft loss, defined as return to maintenance hemodialysis, re-transplantation or listing for re-transplantation, or record of graft failure in the SRTR database. Death with functioning graft was considered a positive outcome. We used Kaplan-Meier [25] survival estimates analyses to evaluate median graft survival in the reference cohort data. Statistical analyses were performed in SAS.

3. Results

3.1. Transplant candidates and donors

We performed an optimized allocation simulation for UTMC transplant candidates and donors procured at Organ Procurement Organizations (OPOs) within 250-nautical miles (Fig. 1). Censored on March 1, 2022, the SRTR database contained a record of 3304 patients waitlisted at UTMC. Out of them, 672 patients were active between January 1, 2014, and December 31, 2019, and these patients were included in the study. The 250-nautical miles around UTMC contained seven OPOs which processed a total number of 6558 deceased donors who had been worked up between the beginning of 2014 and the end of 2019.

Our new algorithm, maximizing matching of kidney transplants by molecular HLA immunogenicity quantification [22], was modified to obtain the least immunogenic donors for patients at UTMC. Based on the availability of 6557 donors, each of 672 patients was evaluated by HMS values against all eligible donors to create 672 donor-recipient pairs with the least possible immunogenic values: each donor was used only once and then automatically eliminated from further search. All inputs for random donor-recipient pairs always observed the requirement of the ABO blood group compatibility.

Basic demographics and characteristics of simulated and reference cohorts are shown in Table 1. Both cohorts were identical with gender and ABO blood group distribution among recipients and donors. Notably, among simulated transplants 45.9 % of recipients had blood type O compared to 43.6 % in real-life transplants (Table 1), albeit the difference was not significant ($p = 0.213$). However, there were more Caucasian and fewer Hispanic recipients, as well as more older patients in the simulated cohort, which may be attributable to the local demographics (Table 1).

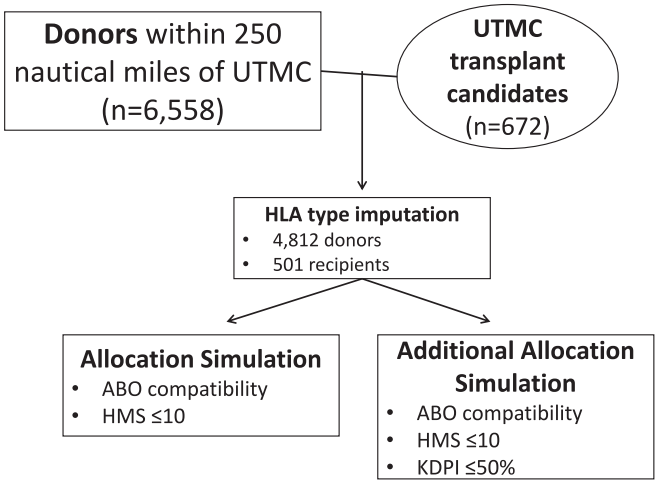


Fig. 1. Diagram explaining the analysis performed in our study. There were 6558 potential donors within a 250-nautical mile area around our Toledo transplant center for 672 transplant candidates active in 2014–2019 at UTMC. An imputation process reduced number of donors to 4812 and 501 recipients based on available HLA-A/B/DR/DQ antigens. Allocation simulations were performed for ABO-compatible pairs to generate transplants with HMS ≤ 10 (first analysis) and ABO-compatible transplants with HMS ≤10 and KDPI ≤50 %.

Table 1
Basic demographics of patients and donors in simulated transplants and the reference retrospective cohort.

	Simulated transplants	Reference cohort	*p*-value
Gender			
Male	304 (61.8 %)	16,925 (61.3 %)	0.8349
Female	188 (38.2 %)	10,673 (38.7 %)	
Blood type			
A	178 (36.2 %)	10,620 (38.5 %)	
O	226 (45.9 %)	12,032 (43.6 %)	0.213
B	66 (13.4 %)	3465 (12.6 %)	
AB	22 (4.5 %)	1481 (5.3 %)	
Race			
Caucasian	298 (60.6 %)	13,703 (49.7 %)	
African American	147 (29.9 %)	9134 (33.1 %)	<0.0001
Hispanic/Latino	22 (4.5 %)	3122 (11.3 %)	
Asian	25 (5.1 %)	1333 (4.8 %)	
Other	0	306 (1.1 %)	
Patient age[a]			
Under 18	0	257 (0.9 %)	
18 to 34	49 (10.0 %)	3911 (14.2 %)	<0.0001
35 to 49	117 (23.8 %)	8075 (29.3 %)	
50 to 65	188 (38.2 %)	11,090 (40.2 %)	
Over 65	138 (28.1 %)	4265 (15.4 %)	
Pre-transplant dialysis			
Yes	412 (83.7 %)	20,174 (73.1 %)	<0.0001
No	80 (16.3 %)	7396 (26.8 %)	
Unknown	0	28 (0.1 %)	
Work for income			
Yes	177 (36.0 %)	3796 (13.8 %)	<0.0001
No	311 (63.2 %)	8756 (31.7 %)	
Unknown	4 (0.8 %)	15,046 (54.5 %)	
Private insurance			
Yes	239 (48.6 %)	11,096 (40.2 %)	<0.0001
No	253 (51.4 %)	16,502 (59.8 %)	
Donor age			
Under 18	51 (10.4 %)	3062 (11.1 %)	
18 to 34	172 (35.0 %)	7640 (27.7 %)	0.0062
35 to 49	125 (25.4 %)	8535 (30.9 %)	
50 to 65	125 (25.4 %)	7251 (26.3 %)	
Over 65	19 (3.8 %)	1110 (4.0 %)	
Donor gender			
Male	286 (58.1 %)	16,085 (58.3 %)	0.9456
Female	206 (41.9 %)	11,513 (41.7 %)	
Donor race			
Caucasian	355 (72.2 %)	22,320 (80.9 %)	
African American	117 (23.8 %)	4160 (15.1 %)	<0.0001
Hispanic/Latino	15 (3.0 %)	414 (1.5 %)	
Asian	4 (0.8 %)	563 (2.0 %)	
Other	1 (0.2 %)	141 (0.5 %)	
Donor ABO			
A	102 (20.7 %)	10,661 (38.7 %)	
O	362 (73.6 %)	12,835 (46.5 %)	<0.0001
B	28 (5.7 %)	3154 (11.4 %)	
AB	0	948 (3.4 %)	

[a] For the reference cohort, patient age is the age at transplant, and for the simulated cohort, patient age is the age at activation on waiting list.

3.2. Matching efficiency of all transplant candidates to donors from the local area

Our newly developed algorithm was applied to 6558 donors to find the best low HMS immunogenic matches for 672 recipients in a single iteration of matching allocation. The number of donors and recipients was reduced to 4812 and 501, respectively, because some HLA haplotypes could not be converted to high-resolution types. Two simulation sets were run: first, the initial simulation where ABO compatibility and optimized HMS value was required; and second, where ABO compatibility and optimized HMS wree required in the additional of KDPI ≤50

%.

In the initial simulation, 472 out of 501 transplant candidates were successfully matched with the HMS threshold of 10, amounting to the matching efficiency of 94.2 %. In the additional simulation, out of 501 transplant candidates, 500 patients were matched with the most optimal low immunogenic ABO-compatible kidney transplants where the HMS values were lower than 10, amounting for the matching efficiency of 99.8 %. Thus, nearly the entire patient cohort was matched with donors with HMS values ranged between 0 and 10.

When viewed by specific HMS value brackets, 13.6 % of simulated transplants in the second simulation had HMS value of 0, compared to 6.6 % of retrospective transplants (Fig. 2A). Similarly, antigenic HLA-A/B/DR mismatch values were distributed where 31.4 % of simulated transplants had HLA mismatch value of 0 or 1, compared to 10.5 % of retrospective transplants (Fig. 2B).

In year 2014, there were 98 transplant candidates registered at UTMC and 659 donor kidneys procured in the 250-nautical mile area (Table 2) in the second simulation model. In 2015, there were 85 transplant candidates and 732 donors, and in the year 2019 there were 72 transplant candidates registered and 929 donor kidneys procured. Notably, the minimum number of donors per patient produced by the transplant simulation algorithm was 1 only in a single instance in 2015. In all other cases, allocation simulation produced multiple donors per transplant candidate, with the maximum number of possible donors ranging from 378 to 636 (Table 2).

Published work showed that HLA-DR mismatch is an important measure of transplant immunogenicity due to high clinical relevance of this locus [26,27]. It is therefore possible to reduce the DR mismatch level as a sole indicator of compatibility for transplants [28]. In fact, in the UNOS allocation policy the sole HLA-DR matching is awarded with additional points for priority on the waiting list for certain categories of patients [29]. When HLA-DR HMS values (possible range 0–75) and HLA-DR mismatches (possible 0, 1 or 2) were used as a measure of transplant immunogenicity, the average DR HMS value was 3.90 (median 3.7) for the simulated transplants including 80 % out of 501 patients achieving an average zero HLA-DR mismatch, indicating at an excellent immunological match. Comparison with the real-life cohort showed that recipients were transplanted with donors having a mean DR HMS value of 4.54 (median 4.25), while the mean HLA-DR mismatch was 1.2, thus showing a significant -DR incompatibility ($p < 0.001$).

3.3. Transplant survival among simulated and reference cohorts

The comparison to the immunogenicity level of real-life performed kidney transplants confirmed the efficiency of our allocation simulation algorithm. Indeed, an average HMS score in simulated matches was only 1.41 (median 1.06), while an average HMS value in the real-life cohort was 6.33 (median 6.28, Fig. 1A; $p < 0.001$). Thus, the 2000–2010 real-life reference cohort confirmed the low immunogenicity among choices for the optimally matched cohort (Fig. 2A and B). The distribution of HLA-A/B/DR antigenic mismatches also was radically different between simulated and reference cohorts: the median 2 HLA mismatches for simulated pairs coincided with 4 perfectly matched HLA alleles out of 6 possible HLA-A/B/DR mismatches, whereas the real-life cohort showed 5 out of 6 HLA-A/B/DR mismatches (Fig. 2B). These results show that patients waitlisted at UTMC could have been matched to excellent donors under the new 250-nautical mile program.

Table 2
Donor numbers per transplant candidates generated by in the additional allocation simulation.

Year	Registered transplant candidates	Donors allocated to transplant candidates			Average donors per transplant candidate
		Total	Minimum	Maximum	
2014	98	659	8	378	149 ± 86
2015	85	732	1	480	166 ± 94
2016	92	756	10	363	187 ± 129
2017	66	828	4	619	195 ± 136
2018	82	880	2	562	200 ± 132
2019	72	929	9	614	229 ± 152

Transplant survival normally cannot be predicted for simulated donor-recipient pairs because these pairs have no outcome data. Indirectly, we assessed the transplant survival of simulated transplants by inferring to retrospective transplants from the retrospective SRTR database analysis with similar levels of HMS immunogenicity [6]. The unadjusted population of transplant recipients in the reference cohort had overall graft survival of 13.4 years (Fig. 3A), which was comparable to a median graft survival for all deceased donor transplants reported by the OPTN data analysis [1]. When the HMS threshold–observed in simulated transplants–was applied to the reference cohort with the HMS threshold of 1.41, their median projected graft survival extended to 18.2 years ($p = 0.0004$, Fig. 3B), a gain of 4.9 years, Table 3.

Similar survival analysis was performed with the focus on HLA-DR locus. The median graft survival for patients under the threshold of HLA-DR HMS $\leq$ 3.9 was 13.9 years, while patients with transplants showing an HMS of >3.9 had median survival of 12.5 years ($p < 0.001$, Fig. 3C). When antigenic mismatch of 0, 1, 2 DR mismatches was used as the immunogenicity threshold, the median survival was 15.1 years in the DR mismatch 0 cohort vs. 12.6 years in the DR mismatch >0 cohort ($p < 0.001$, Fig. 3D). These results suggest that the median graft survival

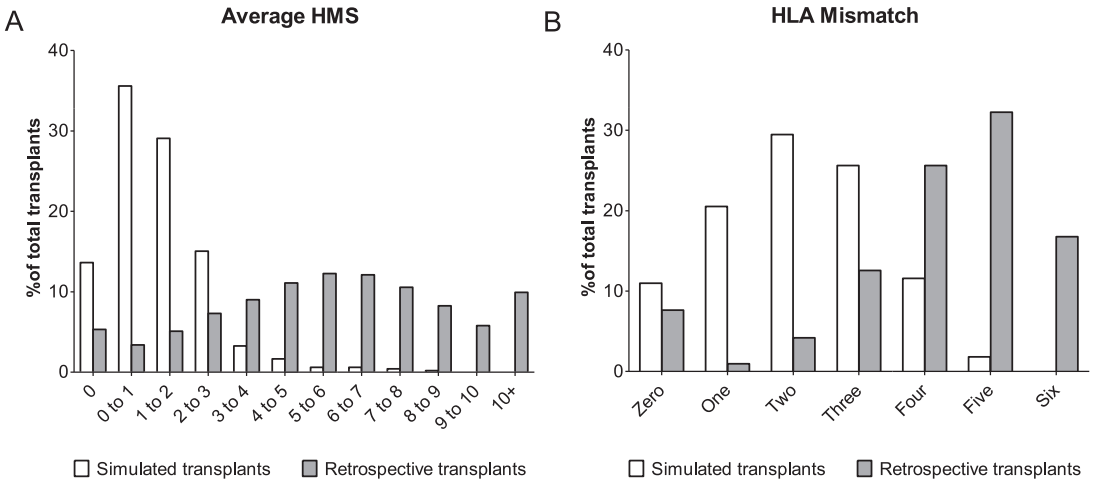


Fig. 2. Immunogenicity of transplants generated in the initial simulation measured using the HMS scale (A), or HLA antigenic mismatch (B) in the simulated transplants and in transplants from the retrospective reference cohort.

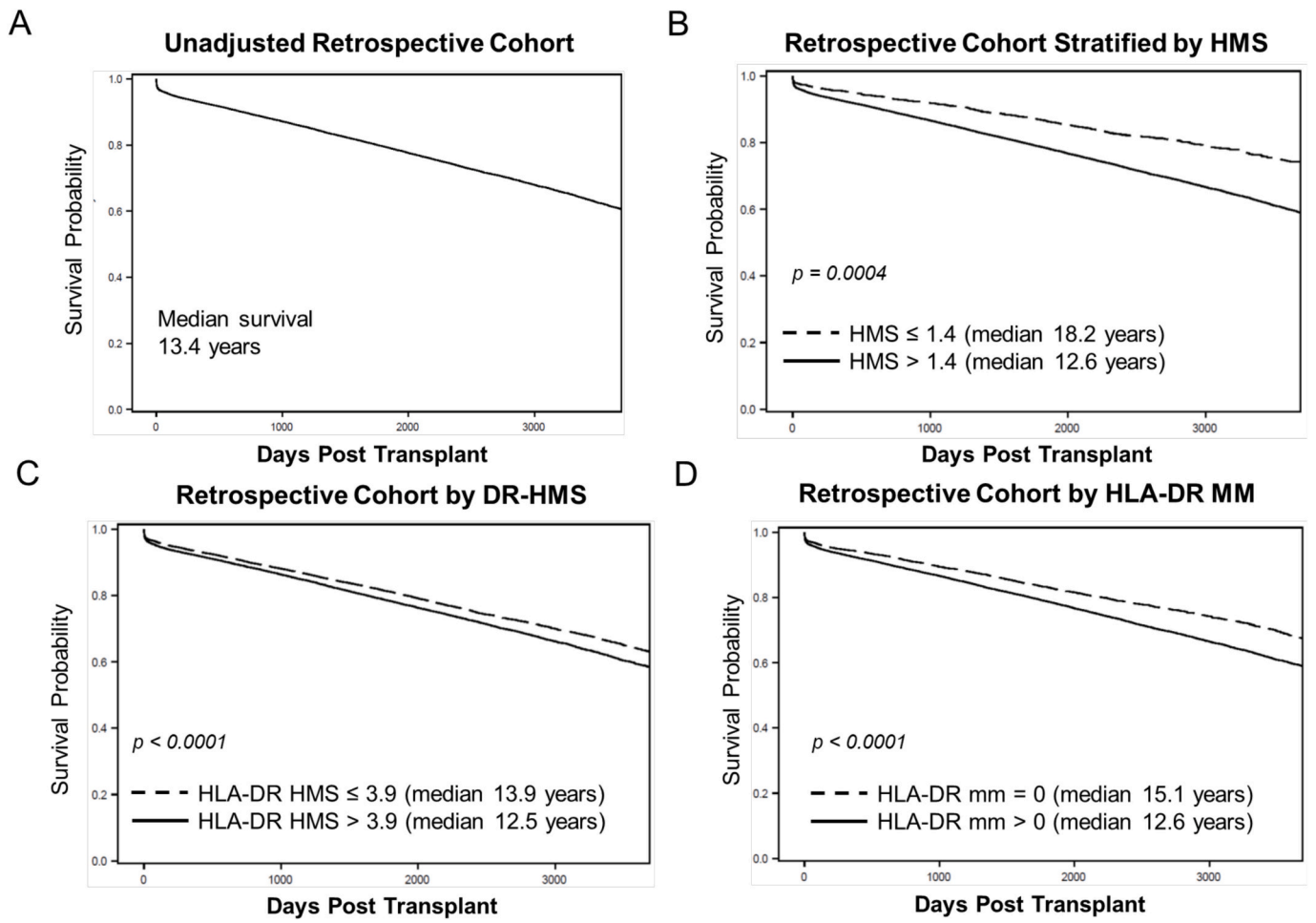


Fig. 3. Death-censored graft survival in retrospective patients categorized by the transplant immunogenicity threshold similar to those observed in simulated transplants. A, Overall survival in unadjusted reference cohort; B, Death-censored graft survival in the reference cohort stratified by HLA-A/B/DR HMS = 1.4; C, Stratification by HLA-DR HMS threshold of 3.9 and D, stratification by HLA-DR antigenic mismatch threshold of 0.

Table 3
Proportion of transplants fitting each immunogenicity category in retrospective patient cohort and the cohort simulated initially.

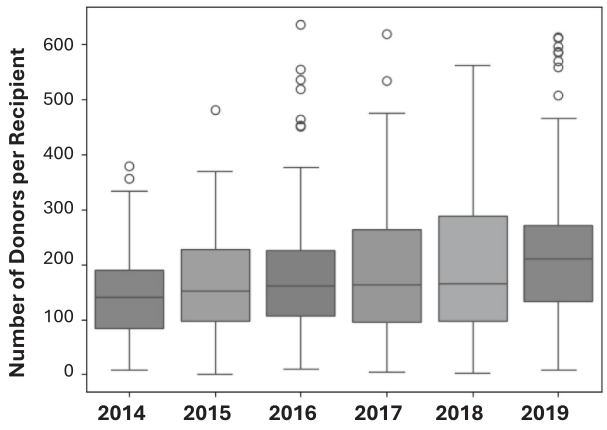
Immunogenicity threshold	Percentage of patients		Associated graft survival
	Retrospective	Simulated	
A/B/DR HMS = 0	6.6 %	13.6 %	18.2
A/B/DR HMS ≤ 1	8.8 %	49.2 %	14.1
A/B/DR HMS ≤ 2	11.4 %	78.3 %	14.1
A/B/DR HMS ≤ 3	15.8 %	93.3 %	14.0
A/B/DR HMS ≤ 4	23.1 %	96.5 %	13.9
A/B/DR HMS ≤ 5	33.7 %	98.2 %	13.8

Death-censored graft survival associated with each immunogenicity category is provided.

in the simulated cohorts was comparable or greater than the median survival in the reference retrospective patient cohort.

3.4. Estimated probability of obtaining multiple low immunogenic donors for each transplant candidate

One of the obvious questions is whether it is possible to find multiple low HLA immunogenic donors for patients at UTMC, which would meet the strict criteria of HMA ≤10 and KDPI ≤50 %. For such analysis we divided the transplant candidate for each year from 2014 to 2019 (Fig. 3). In the same fashion, we included all donors harvested within the 250-nautical mile region for the same time. For 98 transplant candidates for year 2014 we had potential access to 659 donated kidneys. Our simulation revealed that each patient in year 2014 had a median of 141 possibilities for getting low HLA immunogenic kidney with HMS ≤10 and KDPI ≤50 %; patients had a minimum of 8 matching kidneys or a maximum of 378 matching kidneys (Fig. 4). While in the year 2015 one recipient matched only one donor with HMS ≤10/KDPI ≤50 % conditions, another recipient matched 480 out of 732 available donors (Fig. 4); the median was 160 kidneys per recipient. In year 2016, 92 recipients had access to 756 potential ABO-matched donors with a median of 165 kidneys donors with a minimal number of 10 kidneys and maximal of 636 kidneys. Similarly for the year 2017, out of 828 offered kidneys for 66 patients there was a median of 162 kidneys with a minimal number of 4 kidneys and a maximum of 619 kidneys. For the year 2018, out of 880 donors we found 2 potential matches for one patient, but a median was 166 kidneys, while a maximum was 562 kidneys. Finally, in the year 2019 for 72 patients we estimated a median was 210 kidney out of 929 available kidneys with at least 9 kidneys and as many as 614 kidneys (Fig. 4). This analysis confirms our assumption that we can impose very strict requirements (HMS ≤10/KDPI ≤50 %) for low immunogenic donors to be sought for every recipient and find with one exception for 2015 multiple available low immunogenic kidney. These results suggest that it is possible to accommodate most if not all patients with multiple low HLA immunogenic kidneys.



Year	Total Recipients	Total Donors	Minimal donors per recipient	Median donors per recipient	Maximum donors per recipient
2014	98	659	8	141	378
2015	85	732	1	152	480
2016	92	756	10	161	636
2017	66	828	4	162	619
2018	82	880	2	166	562
2019	72	929	9	210	614

Fig. 4. A feasibility analysis of multiple low immunogenic kidneys for each patient. Potential number of low immunogenic kidneys (HMS ≤10/KDPI ≤50 %) was calculated for each patient for individual years from 2014 to 2019. Each year includes all listed transplant candidate for respective year (first column) and the total number of available donor kidneys for each year (second column). The analysis measured a minimal (third column), median (fourth column), and maximum (fifth column) number of kidneys for patients from each year.

4. Discussion

The 2017 lawsuit by a dissatisfied patient, demanding much broader geographical access to kidney donors, triggered an investigation followed by the development of new allocation rules [30]. As a result, the new 250-nautical mile concentric circle allocation policy was implemented in 2018 introducing a drastic change in the nationwide allocation policy. This new rule was developed based on a multitude of allocation simulations performed using the SRTR's Kidney-Pancreas Simulated Allocation Model (KPSAM), in which various distances from 150 to 500 nautical miles were tested [31]. The simulated distances along with feedback from the transplant community instructed the approval for the current 250-nautical mile policy by the OPTN Board of Directors [32]. The intention was to improve the equity of access to donors exemplified by increased of transplant rates for pediatric, female, racial minorities, patients who waited more than 5 years on the waiting list, and highly sensitized patients with the calculated panel of reactive antibodies (cPRA) of 99 % and 100 %.

Current replacement of traditional local service areas with new 250-nautical-mile circles around transplant centers did not change transplantation rates for patients with cPRA less than or equal 98 % [9,31]. Moreover, there was a slight decrease in transplants among nonmetropolitan candidates and those with the Estimated Post-Transplant Survival (EPTS) score less than 20 (scale ranging from 0 to 100) [31]. Another unintended consequence of the new allocation rule was the increase in discarded kidneys, which now is over 20 % nationwide [3,33]. This is because a significant proportion of kidneys was turned down due to reasons unrelated to the organ quality, thus concerns are being raised about organ utilization having negative impact on the entire system design [9,31]. Equally, potentially harmful and unintended consequence of changed policy is a dramatically extended time of kidney preservation to 25 h [9].

In this study, we demonstrated the feasibility for pursuing much better possibilities for HLA matching of kidney transplants with the least HLA immunogenic donors, a clear benefit from the large pool of donors within 250-nautical miles. Our results suggest excellent HLA matches for over 80 % of patients registered at UTMC with an HMS ≤1.4 (range from 0 to 75). Consequently, an average transplant would gain 4.9 years of additional survival and adding 3293 survival years for 762 patients. Previous work by others and our group documented that improved HLA matching correlated for longer graft survivals and less rejection episodes [6,11,34]. Although most transplants are performed with high degree of HLA mismatch, believing that potent immunosuppression overcomes the negative effect of HLA immunogenicity, our analysis should initiate the discussion about new possibilities. For deceased donors, HLA matching presents new challenges with the need for high-resolution HLA typing and the progress in overcoming logistical barriers. Typing techniques are currently being developed with turn-around time sufficiently quick for use in deceased donors, making it feasible to introduce high-resolution typing to deceased donations in near future [35]. Our current analysis showed that the 250-nautical-mile rules created a unique opportunity by the requirement for sharing large number of available donors. Indeed, our pool of eligible 501 patients had access to 4812 donors during the 5-year period.

Our simulation of 501 recipients selected 500 ABO-compatible donors out of 4812 available donors with an average low HMS immunogenicity of 1.4 who were perfectly matched in 4 HLA-A/B/DR alleles. Furthermore, we believe that our HMS-based method would allow us to select additional low immunogenic donors out of such large pool of 4812 donors, suggesting that our approach is fully realistic. Our ongoing research has expanded our model to include transplant centers localized within UTMS' 250-neutical-mile circle, and possibly competing for the same donors. We intend to document that because each center has its own 250-neutical-mile zone, each center has access to a unique pool of donors with the same potential for finding low immunogenic donors. The 2.4 HMS value in simulated cohort was significantly lower than the 5.82 average HMS value of for real-life reference transplants. It is noteworthy to stress that the 2000–2010 reference cohort is partially related to the time when UNOS was assigning points for HLA-B matching [36], thus resulting in overall lower immunogenicity in the cohort. Our recent HLA immunogenicity study observed an average HMS value of 7.8 in transplants from the 2000–2016 cohort [8]. According to the 2020 OPTN report, 83.4 % of deceased donor transplants had 4 or more HLA mismatches [3,5], confirming our current analysis for our reference cohort. The reference cohort had 4 HLA-A/B/DR mismatches vs. only 2 mismatches in the simulated cohort. We propose that the new 250-nautical mile circular requirements create unique opportunities for all transplant centers to reconsider their approach for donors with the low immunogenicity HLA matching. As presented, we selected pairs with low HLA immunogenicity with high HLA-A/B/DR compatibility.

Our simulation showed that every year (2014–2019) the majority of UTMC patients could have multiple low immunogenic donors (HMS ≤10/KDPI ≤50 %). This analysis suggests that even considering the necessity to extend a wait-time to maximum 12 months may be beneficial to find a low immunogenic donor (Fig. 4). Although our 250-neutical mile region includes many transplant centers competing for some donors, because they would get additional low immunogenic offers from their 250-natical mile regions. Our observation of multiple available low immunogenic kidneys for patients supports the feasibility of improving HLA matches for the majority if not all patients.

Although the main limitation of our study was its focus on transplant candidates from a single transplant center, its design documented the possibility of finding low HLA immunogenic donors within 250-

neutical-mile radius. However, we are fully aware that while the feasibility of finding a well-matched donor for each patient is relatively easy to show within a single transplant center, taking the competition from other transplant centers within the 250-nautical mile radius is necessary for complete the full understanding of the donor-recipient matching under the new allocation policy. In addition, our allocation simulation has also not considered the factor of waiting time, as all donor-recipient matching happens simultaneously within the algorithm. Since our analysis included patients who obtained kidney transplants within a 5-year period, this needs to be taken into future consideration in next simulation analysis. These and other additional factors such as patient demographics, cPRA, patient EPTS/donor KDPI scores, as well as offer rejection rates may influence a simulation analysis. Despite these potential shortcomings, our simulation confirms that considerations for improved HLA matching may be the part of improving survival of kidney allografts.

In summary, our first step simulation analysis showed the feasibility for transplant candidates from one center to find better HLA matched donors using the new 250-neutical-mile deceased donor kidney allocation policy. This will become especially important once the high-resolution HLA typing becomes available for all deceased donors and recipients. As exemplified by our pilot analysis of the 4812 available donors within the 250-nautical mile radius from a given transplant center with 501 transplant candidates, transplant immunogenicity may be reduced, and graft survival may be improved to over 18 years for expanded population of deceased donor kidney transplant recipients. The gaining graft survival time would be 4.9 years for every patient with 3293 transplant survival years for all 672 patients included in this analysis.

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CRediT authorship contribution statement

Fayeq Jeelani Syed: Writing – review & editing, Supervision, Methodology, Data curation. **Dulat Bekbolsynov:** Writing – original draft, Conceptualization. **Robert C. Green:** Software, Methodology. **Devinder Kaur:** Supervision, Resources. **Obi Ekwenna:** Funding acquisition, Writing – review & editing. **Puneet Sindhwani:** Methodology, Writing – review & editing. **Michael Rees:** Conceptualization, Funding acquisition, Writing – review & editing. **Stanislaw Stepkowski:** Writing – review & editing, Resources, Investigation.

Declaration of competing interest

None.

Data availability

Data will be made available on request.

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